

# **STA 131A: Introduction to Probability Theory**

## **Lecture 25: The Central Limit Theorem**

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# Announcements

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**Final exam:** Thu, June 11, 1:00–3:00 PM in Wellman Hall 226 (=classroom)

- **Cumulative:** Lecture 1–Lecture 26
- **Arrive early:** The exam starts at 1:00 PM and ends at 3:00 PM sharp.
- **Three hand-written cheat sheets:** Letter-size (8.5" × 11"), double-sided, brief formulas/notes
- **Calculator:** A simple non-graphing scientific calculator is allowed
- **No other materials** beyond the three permitted cheat sheets are allowed (no textbooks, etc.)

**Homework 7** is due tomorrow (Tue, June 2, 11:59 pm)

**Course evaluation:** Please share your feedback comments by Thu, June 4

# Agenda

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## Last time:

- Convergence in probability:

$$Y_n \xrightarrow{P} a \iff P(|Y_n - a| \geq \epsilon) \rightarrow 0, \quad \forall \epsilon > 0$$

- Weak law of large numbers:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{P} \mu$$

## Today: Central limit theorem

- What WLLN tells us, and what it does not tell us
- Statement and intuition of the CLT
- Normal approximation for sums and averages
- De Moivre–Laplace approximation to the binomial

## Recap: What WLLN tells us

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Let  $X_1, X_2, \dots$  be i.i.d. with

$$\mathbb{E}[X_i] = \mu, \quad \text{Var}(X_i) = \sigma^2 < \infty.$$

The sample average is

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

The weak law of large numbers says

$$\bar{X}_n \xrightarrow{P} \mu.$$

That is, for every fixed  $\epsilon > 0$ ,

$$P(|\bar{X}_n - \mu| \geq \epsilon) \rightarrow 0.$$

**Message:** Sample averages stabilize near the mean (consistency).

## What WLLN does not tell us

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The WLLN does not, by itself, describe the approximate distribution of the error

$$\bar{X}_n - \mu.$$

We know

$$\mathbb{E}[\bar{X}_n] = \mu, \quad \text{Var}(\bar{X}_n) = \frac{\sigma^2}{n}.$$

So the typical size of the error is roughly

$$\frac{\sigma}{\sqrt{n}}.$$

This motivates studying the standardized error:

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \quad \text{or equivalently} \quad \frac{S_n - n\mu}{\sigma\sqrt{n}} \quad \text{where} \quad S_n = \sum_{i=1}^n X_i$$

**Question:** What does this standardized error look like for large  $n$ ?

# Central limit theorem

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## Theorem (Central limit theorem)

Let  $X_1, X_2, \dots$  be i.i.d. random variables with

$$\mathbb{E}[X_i] = \mu, \quad \text{Var}(X_i) = \sigma^2 < \infty, \quad \sigma > 0.$$

Let

$$S_n = X_1 + \dots + X_n.$$

Then, for every  $z \in \mathbb{R}$ ,

$$P\left(\frac{S_n - n\mu}{\sigma\sqrt{n}} \leq z\right) \rightarrow \Phi(z),$$

where  $\Phi$  is the standard normal CDF.

**Key takeaway:** After centering by  $n\mu$  and scaling by  $\sigma\sqrt{n}$ , the sum behaves approximately like a standard normal random variable for large  $n$ .

# Central limit theorem: Interpretation

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The CLT says that for large  $n$ ,

$$P\left(\frac{S_n - n\mu}{\sigma\sqrt{n}} \leq z\right) \approx \Phi(z).$$

Equivalently, for large  $n$ , probabilities involving  $S_n$  can be approximated by

$$P(S_n \leq a) \approx \Phi\left(\frac{a - n\mu}{\sigma\sqrt{n}}\right).$$

For sample averages,

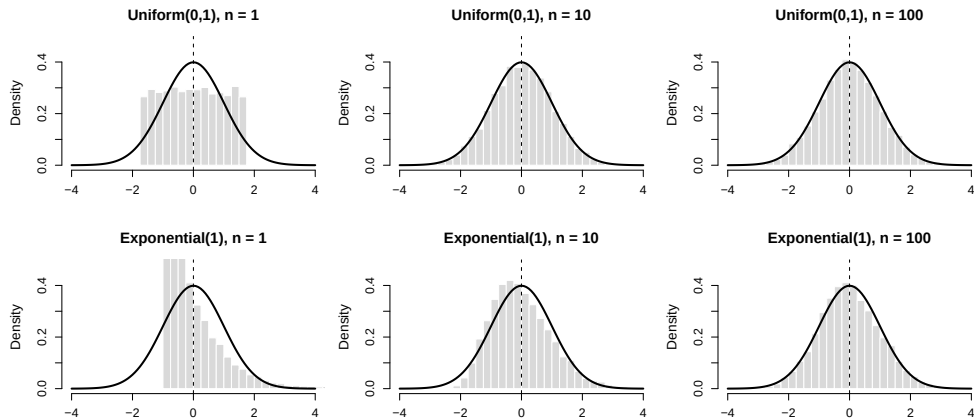
$$P(\bar{X}_n \leq b) \approx \Phi\left(\frac{b - \mu}{\sigma/\sqrt{n}}\right).$$

## Important:

- The individual  $X_i$ 's do not need to be normal.
- Skewness, heavy tails, and discreteness of  $X_i$ 's can affect approximation quality.
- The approximation generally becomes more accurate as  $n$  grows.

# Quality of the normal approximation

## CLT illustration: standardized sample means from 10,000 repetitions



The CLT is asymptotic: accuracy generally improves as  $n$  grows, but the required sample size depends on the shape of the summand distribution.

## Intuition: Why the normal distribution appears

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Assume for simplicity that  $\mathbb{E}[X_i] = 0$  and  $\text{Var}(X_i) = 1$ . Let

$$Z_n = \frac{X_1 + \cdots + X_n}{\sqrt{n}}.$$

If  $M_X(t)$  is the MGF of  $X_i$ , then independence gives

$$M_{Z_n}(t) = \left( M_X \left( \frac{t}{\sqrt{n}} \right) \right)^n.$$

Because  $\mathbb{E}[X_i] = 0$  and  $\text{Var}(X_i) = 1$ , a Taylor expansion around 0 gives

$$M_X(u) = 1 + \frac{u^2}{2} + o(u^2).$$

Thus,

$$M_{Z_n}(t) \approx \left( 1 + \frac{t^2}{2n} \right)^n \rightarrow e^{t^2/2}.$$

Recall  $e^{t^2/2}$  is the MGF of  $N(0, 1)$ . This helps explain why the standard normal appears.

# Why the normal distribution is important

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The CLT helps explain why normal distributions appear so often.

Many quantities arise as sums of many small independent contributions:

$$S_n = X_1 + \cdots + X_n.$$

Even if the individual pieces  $X_i$  are not normally distributed, probabilities involving the standardized sum are often well approximated by standard normal probabilities:

$$\frac{S_n - n\mu}{\sigma\sqrt{n}} \rightarrow N(0, 1) \quad \text{in distribution as } n \rightarrow \infty.$$

**Big picture:**

many independent small effects  $\implies$  approximately normal aggregate.

This underlies many statistical procedures based on standard errors and normal approximations.

# Why study WLLN and CLT separately?

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The WLLN and CLT answer different questions.

## WLLN: stability of averages

$$\bar{X}_n \xrightarrow{P} \mu.$$

- Explains why sample averages and sample proportions are reliable.
- Gives the basic consistency principle:

large sample average  $\approx$  true mean.

## CLT: shape of the remaining fluctuation

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \approx N(0, 1).$$

- Explains the scale of the error:  $1/\sqrt{n}$ .
- Gives normal approximations, standard errors, and margins of error.

**Takeaway:** WLLN says averages stabilize; CLT describes the remaining randomness.

## WLLN vs. CLT

|              | WLLN                                       | CLT                                        |
|--------------|--------------------------------------------|--------------------------------------------|
| Main object  | $\bar{X}_n$                                | $\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}}$  |
| Main message | $\bar{X}_n \rightarrow \mu$ in probability | standardized error is approximately normal |
| Scale        | unscaled error goes to 0                   | error is of order $1/\sqrt{n}$             |

### Interpretation:

- WLLN explains why averages stabilize.
- CLT explains how the remaining error fluctuates.

## Pop-up quiz: What does the CLT say?

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Let  $X_1, \dots, X_n$  be i.i.d. with mean  $\mu$  and variance  $\sigma^2 > 0$ .

**Question:** Which statement is closest to the CLT?

- A) The individual  $X_i$ 's become normal as  $n$  grows.
- B) The sample average  $\bar{X}_n$  is approximately  $N(0, 1)$ .
- C) The standardized average  $\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}}$  is approximately  $N(0, 1)$  for large  $n$ .
- D) The sample average  $\bar{X}_n$  equals  $\mu$  for sufficiently large  $n$ .

**Answer: C.**

The CLT is about the centered and scaled sum or average, not the individual summands and not the unstandardized average.

**Follow-up:** What factors might affect the quality of the normal approximation?

# Normal approximation for sums and averages

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Suppose  $X_1, \dots, X_n$  are i.i.d. with mean  $\mu$  and variance  $\sigma^2$ , with  $\sigma > 0$ .

For the sum  $S_n = X_1 + \dots + X_n$ , the CLT suggests

$$P(S_n \leq c) \approx \Phi\left(\frac{c - n\mu}{\sigma\sqrt{n}}\right).$$

For the sample average  $\bar{X}_n = S_n/n$ , the CLT suggests

$$P(\bar{X}_n \leq a) \approx \Phi\left(\frac{a - \mu}{\sigma/\sqrt{n}}\right).$$

## Workflow:

1. Identify  $\mu$ ,  $\sigma^2$ , and  $n$ .
2. Center and scale.
3. Convert the probability into a standard normal probability.

## Example: Total package weight

### Example

Suppose a plane carries 100 packages whose weights are independent and uniformly distributed between 5 and 50 pounds.

**Question:** What is the approximate probability that the total weight exceeds 3000 pounds?

For a single package  $X_i \sim \text{Uniform}(5, 50)$ ,

$$\mu = \mathbb{E}[X_i] = \frac{5 + 50}{2} = 27.5, \quad \text{and} \quad \sigma^2 = \text{Var}(X_i) = \frac{(50 - 5)^2}{12} = \frac{2025}{12} = 168.75.$$

Let  $S_{100} = X_1 + \cdots + X_{100}$ . Then

$$S_{100} \approx N(100 \cdot 27.5, 100 \cdot 168.75).$$

Thus,

$$P(S_{100} > 3000) \approx 1 - \Phi\left(\frac{3000 - 2750}{\sqrt{16875}}\right) = 1 - \Phi(1.92) \approx 0.027.$$

## Example: Polling error and sample size (1/3)

### Example (Polling)

We poll  $n$  voters and record the fraction  $M_n$  who support a candidate. If  $p$  is the true support probability, write

$$M_n = \frac{X_1 + \cdots + X_n}{n},$$

where the  $X_i$ 's are independent Bernoulli( $p$ ) random variables. Then

$$\mathbb{E}[M_n] = p, \quad \text{Var}(M_n) = \frac{p(1-p)}{n}.$$

By the CLT,  $M_n$  is approximately distributed per normal  $N\left(p, \frac{p(1-p)}{n}\right)$ .

Therefore,

$$P(|M_n - p| \geq \epsilon) \approx 2 \left[ 1 - \Phi \left( \frac{\epsilon}{\sqrt{p(1-p)/n}} \right) \right] = 2 \left[ 1 - \Phi \left( \frac{\epsilon\sqrt{n}}{\sqrt{p(1-p)}} \right) \right].$$

## Example: Polling error and sample size (2/3)

### Example (Polling, continued)

For example, take

$$p = \frac{1}{2}, \quad n = 100, \quad \epsilon = 0.1.$$

Then

$$\frac{\epsilon\sqrt{n}}{\sqrt{p(1-p)}} = \frac{0.1\sqrt{100}}{\sqrt{1/4}} = 2.$$

Thus,

$$P(|M_n - p| \geq 0.1) \approx 2(1 - \Phi(2)) \approx 0.046.$$

By contrast, Chebyshev's inequality gives

$$P(|M_n - p| \geq 0.1) \leq \frac{0.25}{100(0.1)^2} = 0.25.$$

**Message:** Chebyshev is robust but conservative; the CLT often gives sharper approximations.

## Example: Polling error and sample size (3/3)

### Example (Polling sample size)

**Question:** How large should  $n$  be so that

$$P(|M_n - p| \leq 0.01) \approx 0.95?$$

Using the CLT, we want approximately

$$\frac{0.01\sqrt{n}}{\sqrt{p(1-p)}} \geq 1.96.$$

If  $p$  is unknown, use the worst-case bound  $p(1-p) \leq 1/4$ . Then it is sufficient to require

$$\frac{0.01\sqrt{n}}{1/2} \geq 1.96 \quad \implies \quad n \geq 9604.$$

For comparison, using Chebyshev's inequality would give a more conservative conclusion:

$$\frac{1}{4n(0.01)^2} \leq 0.05, \quad \implies \quad n \geq 50000.$$

## Pop-up quiz: CLT standardization

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Let  $X_1, \dots, X_{36}$  be i.i.d. with  $\mathbb{E}[X_i] = 10$  and  $\text{Var}(X_i) = 9$ . Let

$$\bar{X}_{36} = \frac{1}{36} \sum_{i=1}^{36} X_i.$$

**Question:** Which CLT approximation is correct?

- A)  $P(\bar{X}_{36} > 11) \approx 1 - \Phi(1/3)$
- B)  $P(\bar{X}_{36} > 11) \approx 1 - \Phi(2)$
- C)  $P(\bar{X}_{36} > 11) \approx \Phi(2)$
- D)  $P(\bar{X}_{36} > 11) \approx 1 - \Phi(6)$

**Answer: B.** Here

$$\text{SD}(\bar{X}_{36}) = \frac{3}{\sqrt{36}} = \frac{1}{2}.$$

Thus

$$P(\bar{X}_{36} > 11) \approx P\left(Z > \frac{11 - 10}{1/2}\right) = 1 - \Phi(2).$$

# De Moivre–Laplace approximation to the binomial

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We consider a special case of CLT. Let

$$S_n \sim \text{Binomial}(n, p).$$

We can write

$$S_n = X_1 + \cdots + X_n,$$

where  $X_i \sim \text{Bernoulli}(p)$  independently.

Since

$$\mathbb{E}[X_i] = p, \quad \text{Var}(X_i) = p(1 - p),$$

the CLT gives the normal approximation

$$P(S_n \leq c) \approx \Phi \left( \frac{c - np}{\sqrt{np(1 - p)}} \right).$$

This is called the **de Moivre–Laplace normal approximation** to the binomial.

## Continuity correction

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A binomial random variable is discrete, but the normal approximation is continuous.

To approximate probabilities involving integer values, use a **continuity correction**:

$$P(k \leq S_n \leq \ell) \approx P(k - 0.5 \leq G \leq \ell + 0.5),$$

where

$$G \sim N(np, np(1 - p)).$$

For example,

$$P(10 \leq S_n \leq 20) \approx P(9.5 \leq G \leq 20.5).$$

For one-sided events:

$$P(S_n \leq k) \approx P(G \leq k + 0.5), \quad P(S_n \geq k) \approx P(G \geq k - 0.5).$$

**Message:** the adjustment by 0.5 better aligns discrete bars with continuous area.

## Example: Binomial normal approximation (1/2)

### Example

Let  $S \sim \text{Binomial}(100, 0.4)$ .

**Question:** Approximate  $P(35 \leq S \leq 45)$ .

Since  $S$  is a sum of 100 independent Bernoulli(0.4) random variables,

$$\mathbb{E}[S] = np = 100(0.4) = 40, \quad \text{Var}(S) = np(1 - p) = 100(0.4)(0.6) = 24.$$

Thus, for probability calculations, we use  $G \sim N(40, 24)$  as a normal approximation to  $S$ .

Without continuity correction,  $P(35 \leq S \leq 45) \approx P(35 \leq G \leq 45)$ . Therefore,

$$\begin{aligned} P(35 \leq S \leq 45) &\approx \Phi\left(\frac{45 - 40}{\sqrt{24}}\right) - \Phi\left(\frac{35 - 40}{\sqrt{24}}\right) \\ &= \Phi(1.02) - \Phi(-1.02) \\ &= 2\Phi(1.02) - 1 \approx 2(0.8461) - 1 = 0.6922. \end{aligned}$$

## Example: Binomial normal approximation (2/2)

### Example

With continuity correction, the integer event  $35 \leq S \leq 45$  is approximated by the continuous interval  $34.5 \leq G \leq 45.5$ , where  $G \sim N(40, 24)$ .

$$\begin{aligned} P(35 \leq S \leq 45) &\approx P(34.5 \leq G \leq 45.5) \\ &= \Phi\left(\frac{45.5 - 40}{\sqrt{24}}\right) - \Phi\left(\frac{34.5 - 40}{\sqrt{24}}\right) = \Phi(1.12) - \Phi(-1.12) \\ &= 2\Phi(1.12) - 1 \approx 2(0.8686) - 1 = 0.7372. \end{aligned}$$

For comparison, the exact binomial probability is

$$P(35 \leq S \leq 45) = \sum_{k=35}^{45} \binom{100}{k} 0.4^k 0.6^{100-k} \approx 0.7386.$$

**Takeaway:** The continuity correction improves the accuracy of normal approximation.

## Pop-up quiz: Binomial normal approximation

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Let

$$S \sim \text{Binomial}(100, 0.4).$$

**Question:** With continuity correction, which expression approximates  $P(S \leq 45)$ ?

- A)  $\Phi\left(\frac{45 - 40}{\sqrt{24}}\right)$
- B)  $\Phi\left(\frac{45.5 - 40}{\sqrt{24}}\right)$
- C)  $1 - \Phi\left(\frac{45.5 - 40}{\sqrt{24}}\right)$
- D)  $\Phi\left(\frac{45 - 40}{24}\right)$

**Answer: B.**

The event  $S \leq 45$  is approximated by the continuous event  $Z \leq 45.5$ , where  $Z \sim N(40, 24)$ .

**Follow-up:** What continuity correction would you use for  $P(S \geq 45)$ ?

# Wrap-up

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## Central limit theorem

$$P\left(\frac{S_n - n\mu}{\sigma\sqrt{n}} \leq z\right) \approx \Phi(z) \quad \text{for large } n.$$

- CLT describes the approximate shape of the remaining fluctuations.
- Approximation quality depends on  $n$ , distribution shape, and discreteness.

## Normal approximation

- For sums, compare  $S_n$  to  $G_n \sim N(n\mu, n\sigma^2)$  in probability calculations.
- For averages, compare  $\bar{X}_n$  to  $H_n \sim N(\mu, \sigma^2/n)$  in probability calculations.
- Example: Binomial approximation
  - For  $S \sim \text{Binomial}(n, p)$ , use a normal random variable

$$G \sim N(np, np(1-p)).$$

- Use continuity correction when approximating probabilities involving integer cutoffs.

*Suggested reading:* [BT08, Ch. 5.4]

# References

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Dimitri Bertsekas and John N Tsitsiklis.

*Introduction to probability*, volume 1.

Athena Scientific, 2nd edition, 2008.